

Calvin Joyce

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Research Interests

Medical robotics and physical human–robot interaction (pHRI): haptic and bilateral teleoperation, robot learning for contact-rich manipulation, clinical safety formalization, continuum/endovascular robots, and humanoids for clinical tasks, grounded in years of hands-on robot hardware design and deployment.

Seeking a medical robotics R&D / New Product Development internship, available January–September 2027.

Education

University of California, San Diego · B.S. & M.S. Mechanical Engineering, GPA 3.8/4.0

B.S. 2024 · M.S. expected June 2027

Graduate advisor: Prof. Michael C. Yip, Advanced Robotics and Controls Lab (ARCLab)

Research & Work Experience

Advanced Robotics and Controls Lab (ARCLab), UC San Diego

Jan 2023 – Present

R&D Engineer / Graduate Student Researcher · Advisor: Prof. Michael C. Yip

La Jolla, CA

◦ PHABS (Portable Haptic Assisted Bimanual System)

Project lead; first author, ICRA 2027 submission

- * Attacking the force-blindness of video, simulation, and position-only teleoperation data, by capturing force-annotated bimanual demonstrations that anchor imitation-learning corpora.
- * Invented the first handheld bimanual device to render the internal force between two hands on a shared object, the signal that decides crushing, slipping, or tearing, by pairing capstan-driven pinch actuators (maxon RE 36, 27.6 N continuous, 7.94:1 cable transmission) with a gimbal-motor linear squeeze stage on carbon-fiber rails over recirculating-ball bearings.
- * Closed the bilateral teleoperation loop on an Anvil OpenArm dual-arm robot, by estimating contact wrenches from gravity-compensated joint effort (recursive Newton–Euler, damped Jacobian pseudoinverse) and rendering per-hand pinch plus a signed inter-hand internal-force channel.
- * Built the embedded sensing and actuation stack: a Teensy 4.1 fusing IMUs, magnetic encoders, barometers, and time-of-flight ranging over I²C/UART, driving H-bridge and field-oriented-control motor stages, with synchronized device- and robot-frame logging to LeRobot format.
- * Now training ACT policies on matched haptic vs. no-haptic demonstrations across three contact-rich bimanual tasks, with a stiffness-discrimination and NASA-TLX human-subjects study; verified force-rendering fidelity against a reference load cell.

◦ RIC (Robotic Infant Care) — first framework for direct robotic infant handling

Project lead; first author, ICRA 2027 submission

- * Defining safe robotic care for the NICU's most frequent physical interventions: bimanual infant pickup and CPAP nasal-mask repositioning, contact-rich tasks no robotic system has performed on a physical infant.
- * Formalized the safety bar, by deriving continuously measurable clinical thresholds with four NICU clinicians (a NICU nurse, a NICU physician, and two neonatal respiratory therapists): cervical pitch and head acceleration for pickup; contact pressure, eye/mouth exclusion zones, and a PEEP-interruption time budget for CPAP.
- * Benchmarking direct-human, teleoperated, and autonomous (ACT) execution on a Unitree G1 humanoid, by building a REMI-manikin OptiTrack pipeline, molding a Fisher & Paykel nasal mask with embedded force-sensitive resistors and in-situ load-sharing calibration, and verifying every trial against the clinical thresholds.

◦ Humanoid Surgery (*video*) · in vivo feasibility of humanoid robots in laparoscopic surgery

Co-author, *Nature* 2026

- * Enabled humanoid robots to operate standard medical tools in dry-lab and in vivo porcine surgeries, by designing the tool-mount system (optimized mass/CoM, bench- and sim-validated rigidity) and running mechanical setup for every surgical session.
- * Coordinated a 10-person robot–clinician team through live surgeries, turning surgeon feedback into hardware revisions.

◦ Autonomous Catheter Navigation — early-stage ARPA-H & DARPA project; RL for endovascular robotics

Jul 2026 – Present

- * Developing autonomous catheter steering in simulation, by training PPO policies with domain randomization in GPU-physics models of patient-derived vasculature; early policies outperform the project's prior baseline.
- * Made sampling-based MPC tractable in simulation, by building state snapshot/restore for the GPU catheter-physics solver.
- * Built three catheter simulation demos for Newton, an open-source physics engine from NVIDIA, DeepMind, and Disney.

ArtemisAI Labs · Mechanical Engineer, La Jolla, CA

Jan 2026 – Present

- * Made a NICU bedside camera mount clinically deployable, by writing its load, retention, and cleanability requirements (rough-handling loads benchmarked against the IEC 60601-1 instability test; no-disassembly hydrogen-peroxide wipe-down) and redesigning it against them with low-cost, manufacturable, and COTS components.
- * Split the unit into separately specified clamp, arm, and phone-interface subassemblies with acceptance criteria for each (pole capture range, slip torque, pull-off force, reposition-cycle stiffness retention), then wrote the assembly protocol and bill of materials so labor could be quoted independently of parts.
- * Reached the bedside: the mount fields the cameras for NOVA (Neonatal AI Vision Assessment), ArtemisAI's Phase II multi-center trial of computer-vision neurological monitoring of newborns, running at Mount Sinai; lining up assembly partners to kit and ship units to purchasing hospitals.

Research Projects

• Haptic Shoulder (*paper*) · pHRI phantom; co-first author, ICRA 2025; provisional patent

Jan 2024 – June 2024

- * Invented a biomechanically accurate shoulder phantom for safe pHRI testing (provisional patent in progress), by dedicating one motor-encoder module to each anatomical DOF to reproduce joint center, coupled kinematics, and configuration-dependent range of motion.
- * Enabled real-time testing and deployment of pHRI planning tasks onto physical robots without human subjects, standing in for a patient the way a medical phantom does.

• ARCSnake V2 (w/ NASA JPL) (*paper*) · amphibious screw-propelled snake robot; IROS 2026

June 2021 – Sep 2025

- * Increased propulsion-screw output torque 40%, by raising the torque ratio, aligning the belt drive, and cutting friction (shorter drive stack, relieved rubbing surfaces), validated on a testbed I built.

- Characterized which screw parameters drive performance across media, by engineering a mobile testbed for constrained axial measurement with a 6-DOF force-torque sensor, validated via SolidWorks FEA (co-first author, ICRA 2023).
- Designed NASU (Novel Actuating Screw Unit), the first Archimedes-screw locomotion design with a dynamically reconfigurable angle of attack for sand, water, and dirt (first author, ISRR 2024).
- Built the “Voodoo Doll,” a 10-DOF specialized teleoperation controller, by designing lockable U-joints with magnetic encoders to command ARCSnake V2’s joint positions.
- Led hardware integration, electrical and mechanical, for validation of the full robot in underwater testing.

• Autonomous Driving RC Car — course project: vision-based autonomy on embedded hardware

April 2024 – June 2024

- Completed a track autonomously, by implementing edge-detection and line-following in Python with OpenCV on embedded Linux; integrated an IMU to detect hydroplaning and trigger corrective steering.

Publications ([Google Scholar](#))

Presented or co-presented at ICRA 2023, ISRR 2024, ICRA 2025, and ICRA 2026. * denotes equal contribution.

- Z. Liang, N. Thareja, P. Zhang, **C. Joyce**, S. Atar, F. Richter, G. Jacobsen, S. Liu, R. Broderick, and M. C. Yip, “[In vivo feasibility study of humanoid robots in surgery](#),” *Nature*, 2026.
- C. Joyce**, J. Honma, T. Tattala, V. Benedicto, and M. C. Yip, “PHABS (Portable Haptic Assisted Bimanual System) for Force-Annotated Bimanual Shared Object Teleoperated Data,” ICRA 2027 submission, Sept 2026.
- C. Joyce**, S. Hong, R. Fellbaum, F. Richter, and M. C. Yip, “How Safe Is Robotic Infant Handling? Benchmarking Learning from Demonstration Against Clinical Thresholds for Neonatal Pickup and CPAP Mask Repositioning,” ICRA 2027 submission, Sept 2026.
- Z. Liang, X. Liang, S. Atar, S. Das, Z. Chiu, P. Zhang, **C. Joyce**, F. Richter, S. Liu, and M. C. Yip, “[LapSurgie: Humanoid Robots Performing Surgery via Teleoperated Handheld Laparoscopy](#),” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2026.
- S. Wickenhiser*, E. Peiros*, **C. Joyce**, P. Gavrilov, S. Mukherjee, S. Sylvester, J. Zhou, M. Cheung, J. Lim, F. Richter, and M. C. Yip, “[ARCSnake V2: Mechanical Adaptations For An Amphibious Multi-Domain Screw-Propelled Snake-Like Robot](#),” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2026, to appear.
- C. Joyce***, E. Peiros*, T. Murugesan*, R. Nguyen, I. Fiorini, R. Galibut, and M. C. Yip, “[Haptic Shoulder for Rendering Biomechanically Accurate Joint Limits for Human-Robot Physical Interactions](#),” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2025.
- C. Joyce**, J. Lim, R. Nguyen, M. Owens, S. Wickenhiser, E. Peiros, F. Richter, and M. C. Yip, “[NASU — Novel Actuating Screw Unit: Origami-Inspired Screw-Based Propulsion on Mobile Ground Robots](#),” in *Proc. Int. Symp. Robotics Research (ISRR)*, 2024.
- S. Atar, **C. Joyce***, X. Liang*, F. Richter, W. Ricardo, C. Goldberg, P. Suresh, and M. C. Yip, “[Humanoids in Hospitals: A Technical Study of Humanoid Surrogates for Dexterous Medical Interventions](#),” arXiv:2503.12725, 2025. (co-second author)
- C. Joyce***, J. Lim*, E. Peiros, M. Yeoh, P. V. Gavrilov, S. G. Wickenhiser, D. A. Schreiber, F. Richter, and M. C. Yip, “[Mobility Analysis of Screw-Based Locomotion and Propulsion in Various Media](#),” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023.

Patents

- Haptic Shoulder: biomechanically accurate shoulder phantom for pHRI testing. Provisional patent, in progress (co-inventor).
- PHABS: handheld bimanual haptic teleoperation device with inter-hand force rendering. Patent in preparation (lead inventor).
- Helping Hand: catheter-tip end-effector for grasping and cutting. Provisional patent, in progress (co-inventor).

Mentorship

- Mentored 7 students on the humanoid surgery project (concluded): CAD/FEA reviews, DFMA, fixture design, and test plans.
- Mentoring 2 students on PHABS and 2 on RIC toward their ICRA 2027 submissions.
- Mentored 5+ students on ARCSnake V2 through mechanical design reviews, supporting its IROS 2026 paper.
- Coached 20+ mechanical members across Triton Robotics subteams; ran design reviews, fabrication, and assembly for 3 robots.
- Directly mentored 5 members as subteam lead from concept to assembly: requirements, BOM, and manufacturing.

Leadership

Triton Robotics, UC San Diego · Team President / Advisor

June 2020 – June 2023

- Scaled the team 20→80+ members and the proportion of non-male members from 15% to over 35%, by formalizing onboarding and training, DEI partnerships, targeted recruiting, and anonymous feedback loops.
- Led 17 team leads with OKRs, roadmaps, and stand-ups, by founding Business, Marketing, and Project Development teams; produced a formal business plan, sponsorship packet, donation page, and lab fit-out.
- Obtained nationally recognized nonprofit status, leading immediately to \$3K in funding, a corporate sponsorship pipeline, and permanent workspace.

Service & Professional Memberships

- IEEE Robotics and Automation Society (RAS), member since 2023.
- Robotics outreach with Triton Robotics, including events at the UCSD Women’s Center; built DEI partnerships that raised team diversity from 15% to 35%+.

Skills

- Medical Robotics & pHRI:** haptics & bilateral teleoperation, catheter/continuum robots, clinical safety thresholds, OptiTrack
- Software & Control:** Python, C++, MATLAB, PyTorch, ROS 2; imitation learning (ACT), PPO, MPPI; NVIDIA Warp/Newton
- Hardware:** SolidWorks + FEA, Fusion 360, DFMA, resin/FDM printing, laser cutting; design requirements & acceptance criteria, verification testing, test fixtures, capstan drives, motor drivers, sensors

Coursework

Robotics; Haptics; Soft Robotics; Intro to Autonomous Vehicles; Linear Control; Signals and Systems; Mechanical Design; Computational Methods in Design; Experimental Techniques; Mechanical Engineering Lab; Ocean Technology Design & Development; Solid Mechanics; Dynamics and Vibrations; Elements of Materials Science; Fluid Mechanics; Heat Transfer; Linear Circuits; MATLAB for Engineering Analysis; Recommender Systems and Web Mining